

Core Statistics for In Vitro Microbiome Data

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Bioinformatics 101 for colon models experiments



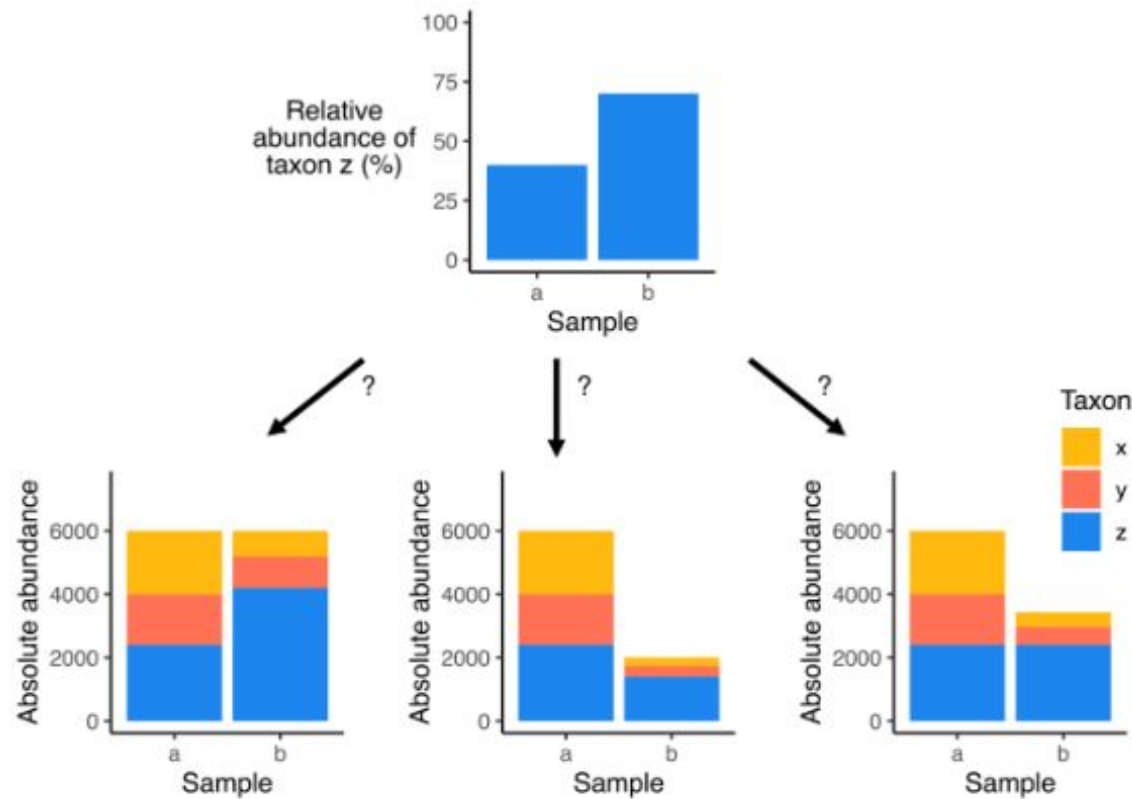
Choosing the right test matters as much as a good study design

“A clean design still gets ruined by the wrong test, or the right test run a few thousand times without correcting for it.”

Session overview

- Exploration of microbiome data
- Hypothesis testing fundamentals
- Testing diversity
- Differential abundance & compositionality
- Multiple comparisons
- Effect size vs. significance

Relative vs absolute abundances



Douglas & Langille, 2021

- Sequencing mostly gives relative abundances that sum to a fixed total, this is called compositional data.
- If one taxon's true absolute abundance rises, every other taxon's relative share must fall to keep the total at 100%.
- Classical statistical methods treat every taxon as independent, which compositional data, by construction, is not.

Gloor et al. (2017), Frontiers in Microbiology

One taxon's real change creates two fake ones

Taxon	Sample 1 (absolute)	Sample 1 (relative)	Sample 2 (absolute)	Sample 2 (relative)
Taxon A (really doubled)	50	50.0%	100	66.7%
Taxon B (unchanged)	30	30.0%	30	20.0%
Taxon C (unchanged)	20	20.0%	20	13.3%
Total	100	100%	150	100%

Taxon B and C's absolute abundance never changed, but their relative abundance dropped by 10 and 6.7 percentage points respectively, purely because Taxon A really doubled. A naive per-taxon test would call all three “significant.”

Controlling for sequencing depth

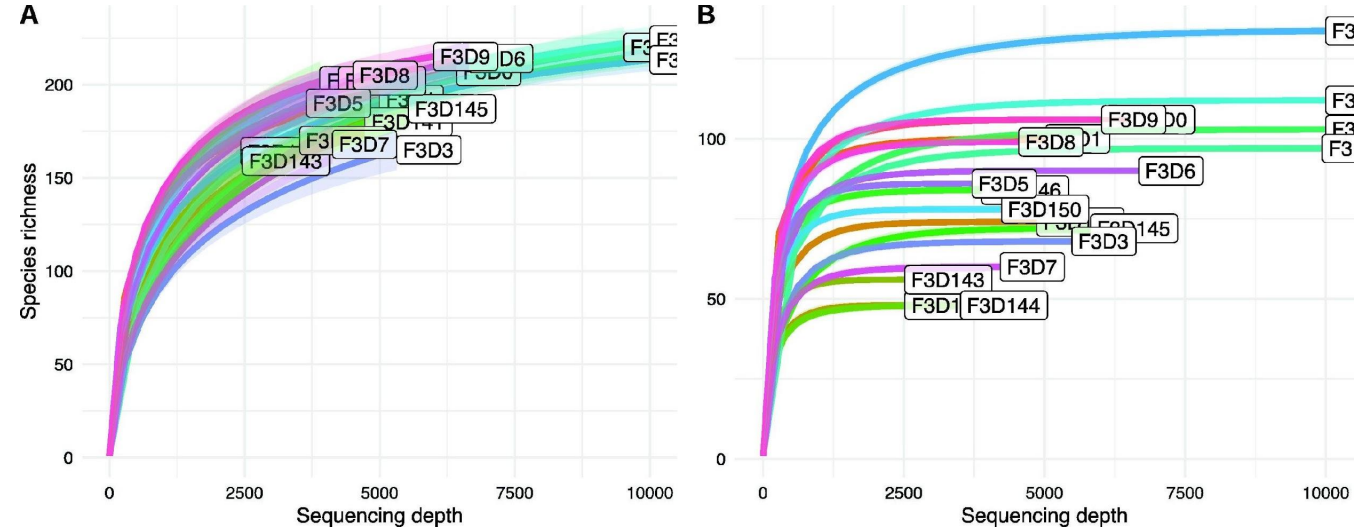
- Sequencing depths varies from sample to sample
 - Not directly related to the original amount of cells/DNA
 - More depth doesn't mean more microbes
- Greater sequencing depth $\square$ greater taxa detection
- We need to control/normalize for sequencing depth

Controlling for sequencing depth

Rarefaction

Subsampling sequencing reads without replacement to a fixed depth so different samples have the same read count.

It's commonly used before **alpha** and **beta diversity** calculations.



Bardenhorst, et al. 2022

Does rarefying solve the problem?

 OPEN ACCESS  PEER-REVIEWED

RESEARCH ARTICLE

Waste Not, Want Not: Why Rarefying Microbiome Data Is Inadmissible

Paul J. McMurdie, Susan Holmes 

Published: April 3, 2014 • <https://doi.org/10.1371/journal.pcbi.1003531>

<https://doi.org/10.1371/journal.pcbi.1003531>

Original Abundance			Rarefied Abundance		
	A	B		A	B
OTU1	62	500	OTU1	62	50
OTU2	38	500	OTU2	38	50
Total	100	1000		100	100

Standard Tests for Difference			
P-value	chi-2	Prop	Fisher
Original	0.0290	0.0290	0.0272
Rarefied	0.1171	0.1171	0.1169

Does rarefying solve the problem?

MINI REVIEW article

Front. Microbiol., 15 November 2017

Sec. Systems Microbiology

Volume 8 - 2017 | <https://doi.org/10.3389/fmicb.2017.02224>

Microbiome Datasets Are Compositional: And This Is Not Optional

Gregory B. Gloor^{1}, Jean M. Macklaim¹, Vera Pawlowsky-Glahn² and Juan J. Egozcue³*

<https://doi.org/10.3389/fmicb.2017.02224>

 | Editor's Pick | Microbial Ecology | Research Article | 6 December 2023



Waste not, want not: revisiting the analysis that called into question the practice of rarefaction

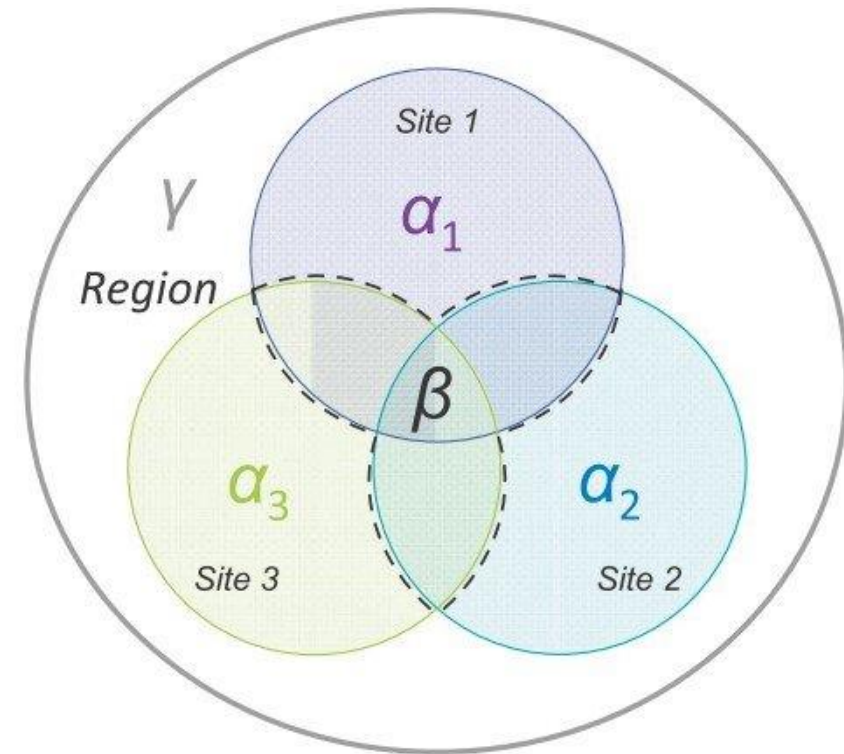
Author: [Patrick D. Schloss](#)   | [AUTHORS INFO & AFFILIATIONS](#)

<https://doi.org/10.1128/msphere.00355-23>

Diversity measures

- Alpha diversity
 - Richness (number of different species),
 - Evenness (whether the species are present in similar even abundances),
 - Shannon index
 - Simpson Diversity
 - Chao1 (predicts the total number of species in a sample based on the rarefaction curve)
- Beta diversity
 - Bray-Curtis index, Jaccard index, Aitchison distance, Unifrac distances
- Gamma diversity

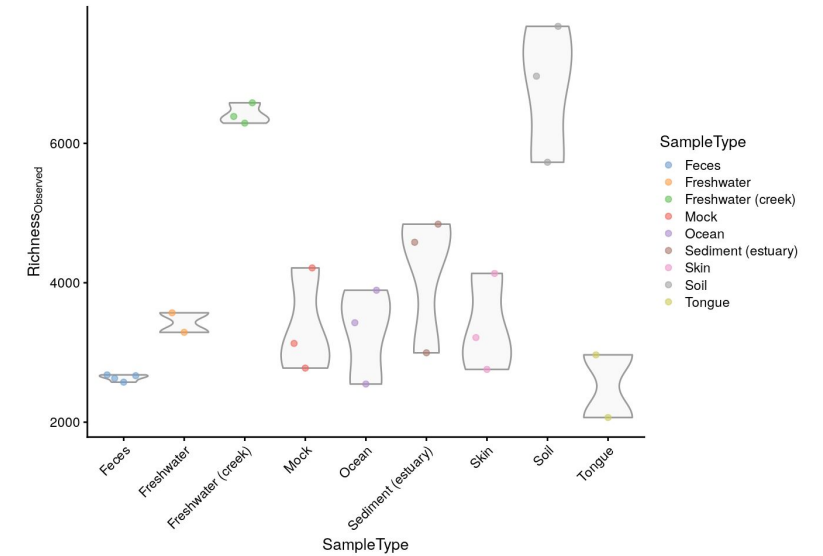
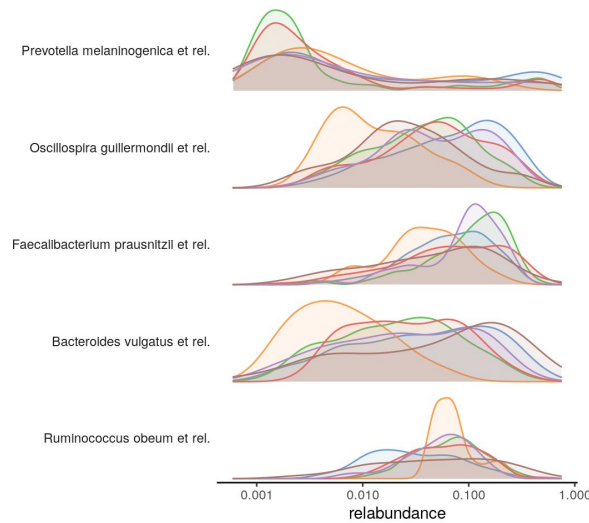
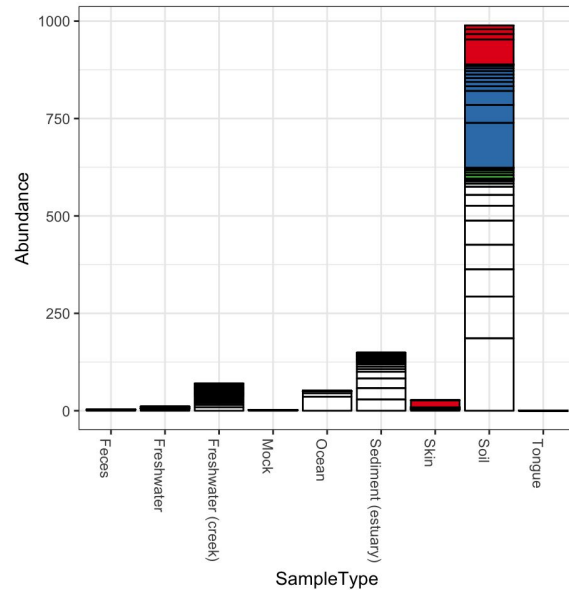
Eestimates diversity within a region



Credit to Jurgens, 2018.

Exploration of microbiome data

Graphical summary, abundance bar, richnees plot.



● Abundance bar

GlobalPatterns data, Phyloseq package

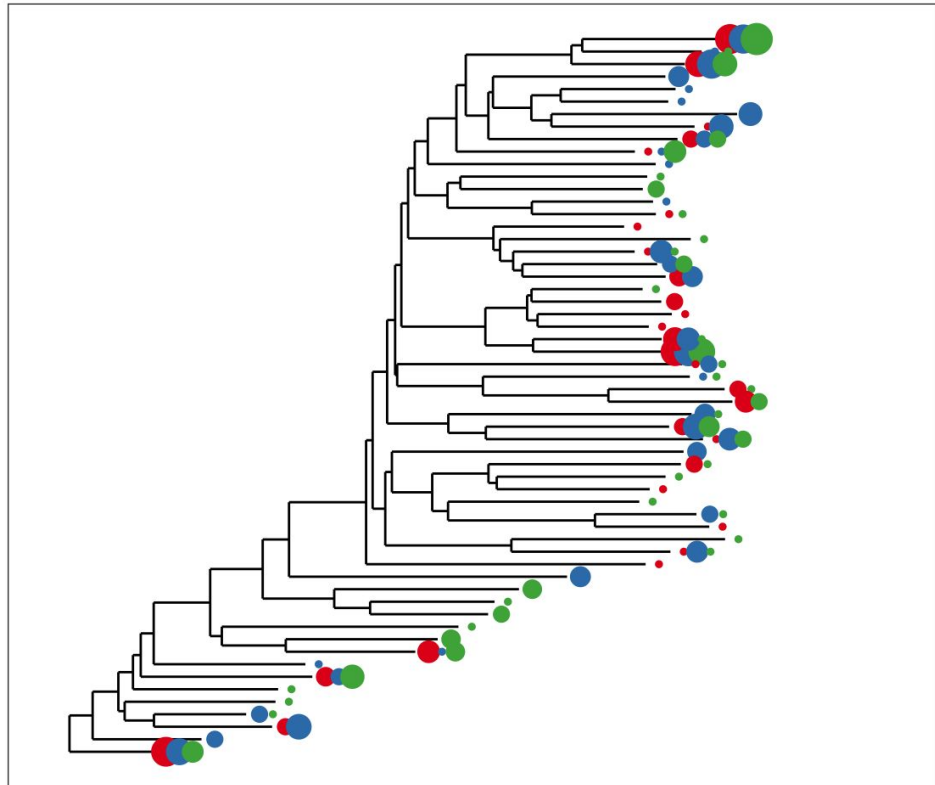
» Richnees plot

GlobalPatterns data, mia package

» Relative abundance

Exploration of microbiome data

Phylogenetic trees.

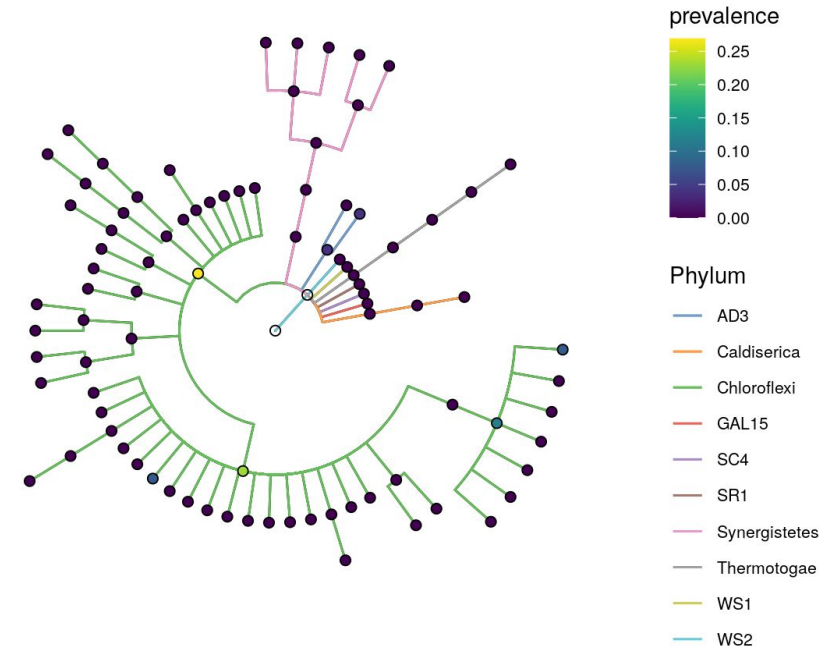


Sample

- B
- C
- D

Abundance

- 1
- 5
- 25



prevalence

- 0.25
- 0.20
- 0.15
- 0.10
- 0.05
- 0.00

Phylum

- AD3
- Caldiserica
- Chloroflexi
- GAL15
- SC4
- SR1
- Synergistetes
- Thermotogae
- WS1
- WS2

» Globalpatterns dataset tree, prevalence of top phyla

- Esophagus dataset tree, [Phyloseq](#)

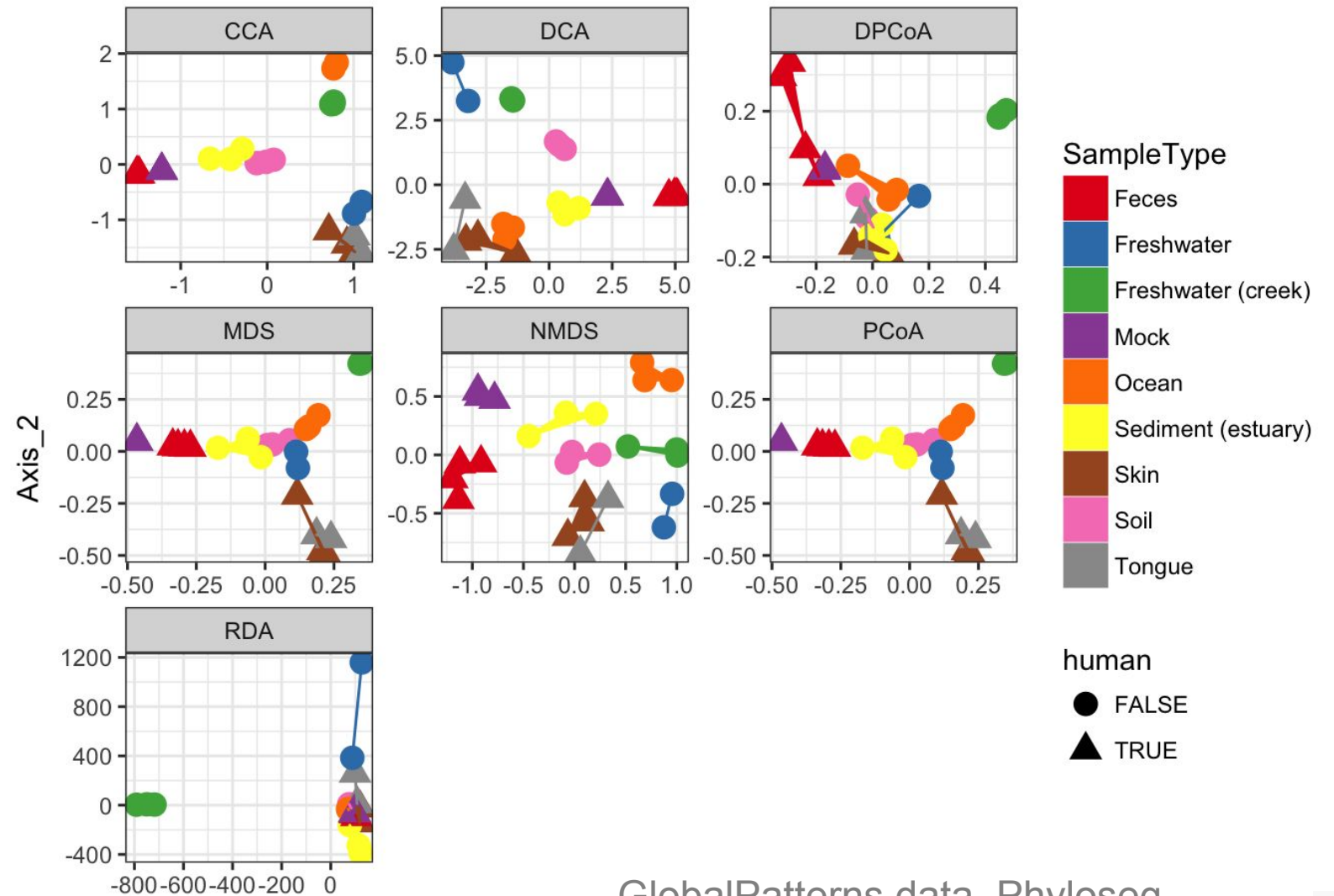
Heatmap and networks.



Exploration of microbiome data

Ordination methods

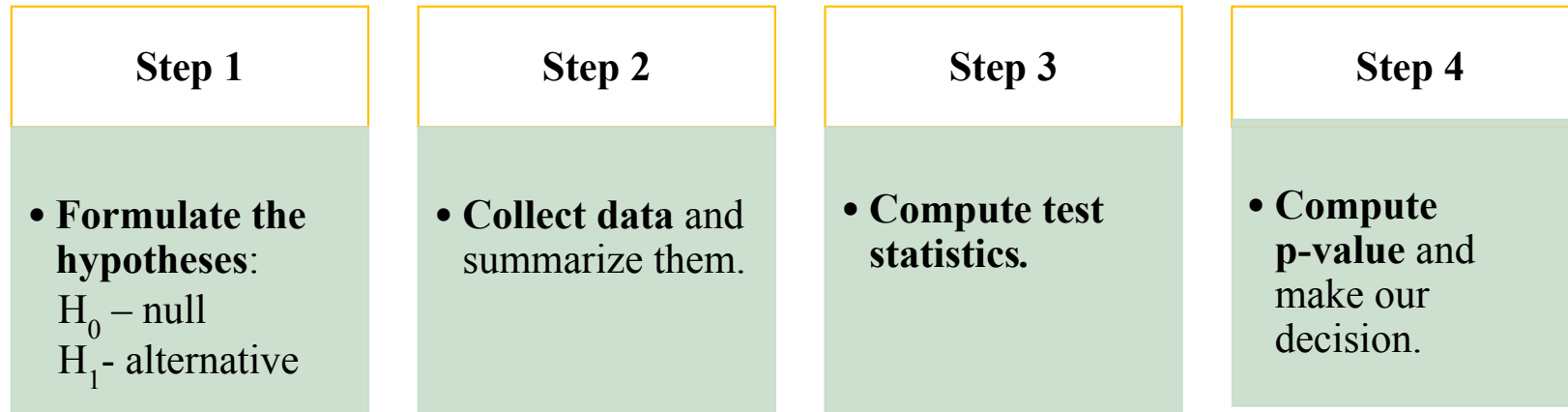
- Canonical Correspondence Analysis (CCA)
- Principal Cordinate Analysis (PCoA)
- Non-metric Multidimensional Scaling (NMDS)
- Redundancy Analysis (RDA)



GlobalPatterns data, Phyloseq

A refresher on hypotheses testing

Statistical hypothesis testing is defined as assessing evidence provided by the data against the null hypothesis.



1. The null hypothesis (H_0) is 'no real difference'.
2. A test statistic (t, U, F, ...) is just a standardized distance between what you observed and what H_0 predicts, the p-value is that distance translated into a probability.

Two ways a test can be wrong

Type I/II error are the α and the β from sample-size calculation.

H_0 true, don't reject

Correct, true negative

H_0 true, reject

Type I error (α), false positive

H_0 false, don't reject

Type II error (β), false negative

H_0 false, reject

Correct, power = $1 - \beta$

Microbiome data usually fails normality and equal-variance checks

Compositional

Sparse

Overdispersed

Non-normal

Parametric tests

Assume normal distribution, equal variance across groups

1. Paired / unpaired t-test (2 groups)
2. One-way ANOVA (3+ groups)

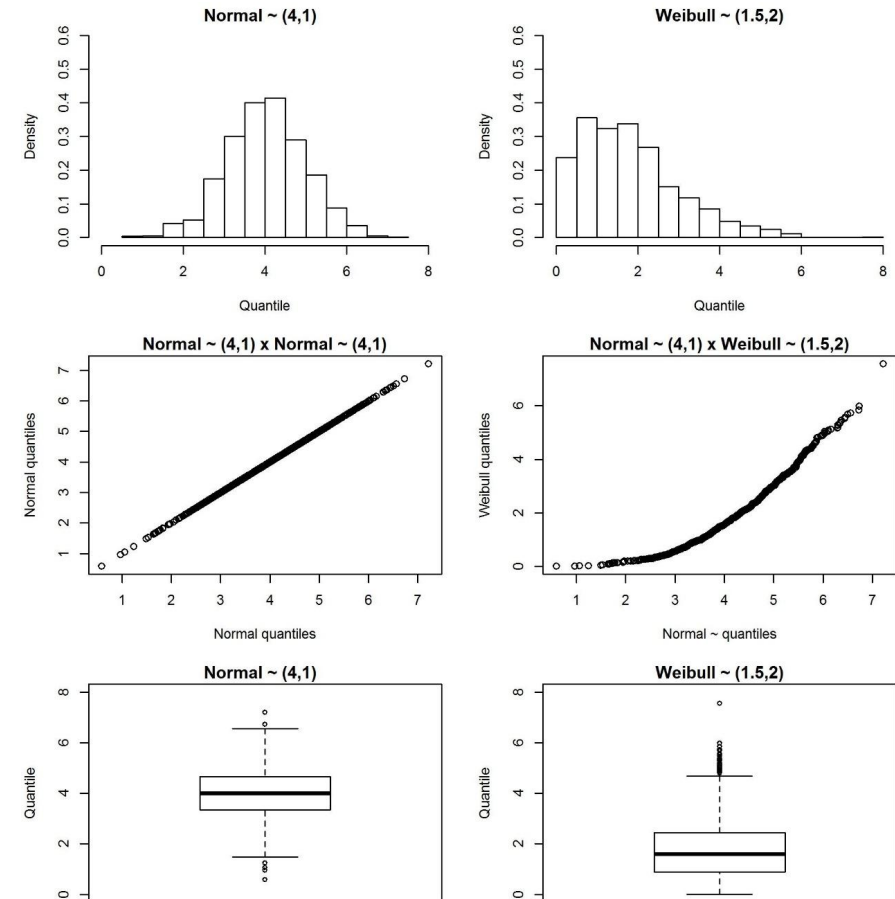
Non-parametric tests

Assume nothing about the shape of the distribution

1. Wilcoxon signed-rank (paired, 2 groups)
2. Mann-Whitney U / Wilcoxon rank-sum (unpaired, 2 groups)
3. Kruskal-Wallis (3+ groups)

Checking assumptions, and letting your design pick the test

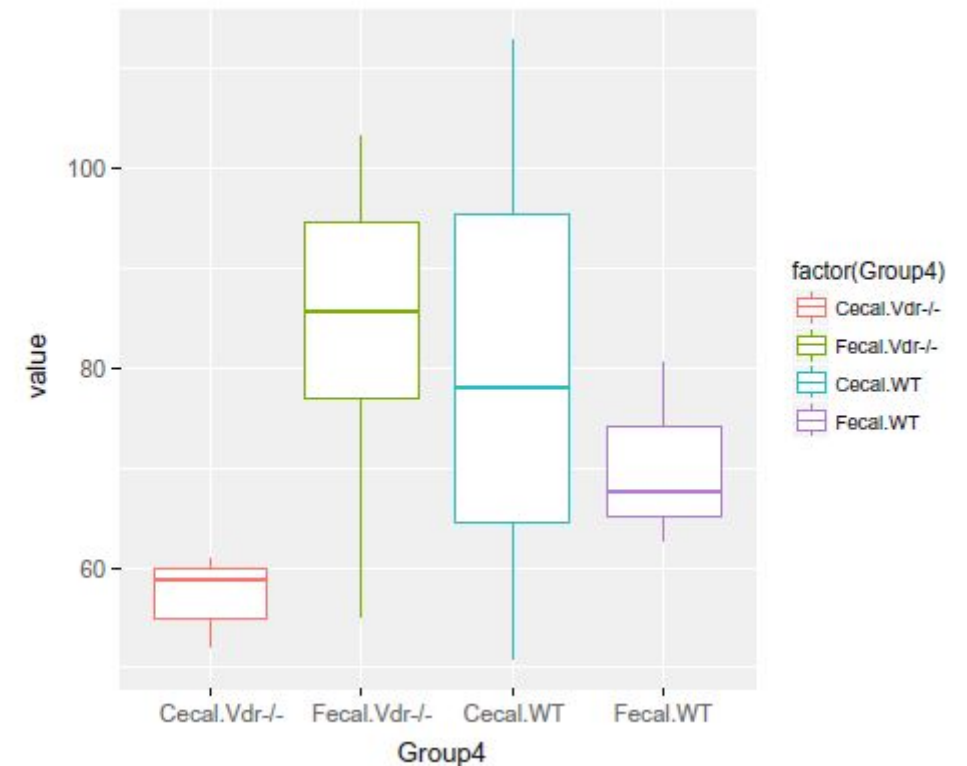
1. Shapiro-Wilk tests normality directly; a quick histogram or Q-Q plot does the same job by eye.
2. Levene's test checks equal variance across groups; unequal variance alone can distort a standard t-test's p-value even when means genuinely differ.



Comparing diversity indices between groups

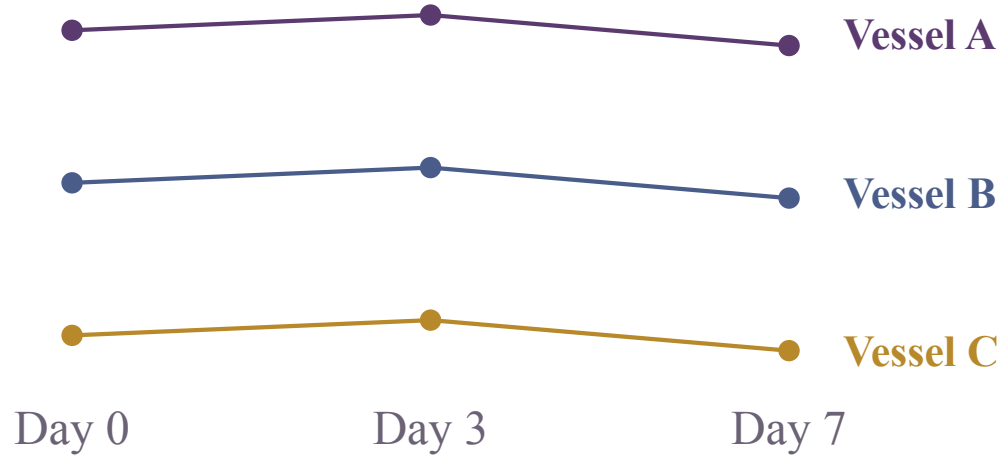
Shannon, Simpson, and richness are single numbers per sample, treat them like any continuous variable, carefully.

1. Alpha diversity indices (Shannon, Simpson, observed richness) collapse a whole community into one number per sample.
2. Independent samples, non-normal or skewed → Wilcoxon rank-sum (2 groups) or Kruskal-Wallis (3+ groups); no distributional assumption required.
3. Report the median difference and a non-parametric effect size alongside the p-value.



-Analysis of Chao 1 alpha diversity measures; Xia et al., 2018. Springer Series in Statistics

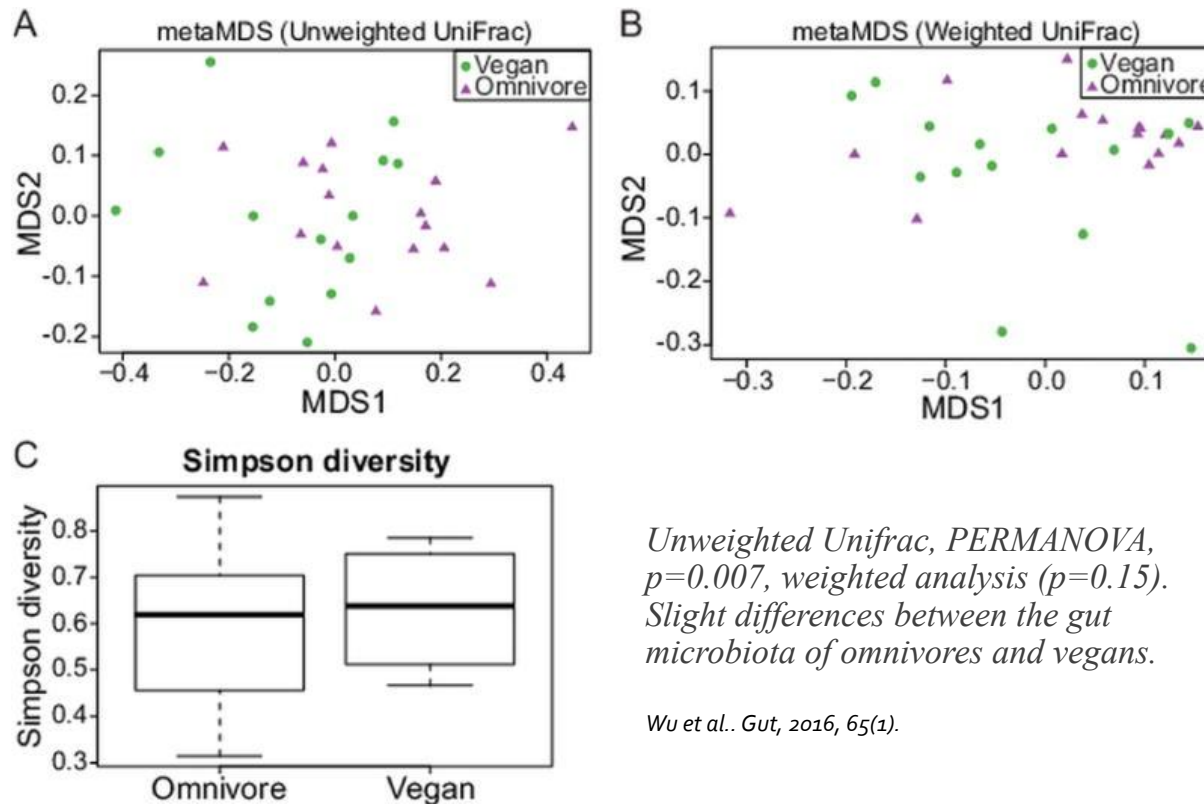
When the same vessel is sampled more than once



1. Treating all 9 points as 9 independent samples overstates your effective sample size and can bias the test.
2. A linear mixed-effects model adds a random intercept per vessel: $\text{diversity} \sim \text{treatment} + (1 \mid \text{vessel})$. The vessel term absorbs each vessel's baseline.
3. A repeated-measures ANOVA is a special case for normally-distributed data; for skewed data, use a mixed model on rank-transformed values.

PERMANOVA tests whole-community differences

Ordination shows you the pattern. PERMANOVA is what actually tests it.



1. Distances between samples come from Bray-Curtis, Jaccard, or UniFrac (a full matrix, not one number).
2. PERMANOVA partitions variance in the distance matrix by group, using permutation rather than a theoretical distribution to get a p-value.
3. Flexible to dissimilarity measure; No assumption of multivariate normality; Not sensitive to differences in correlation structure among groups; Can include random effects, interaction terms, and hierarchical structures.

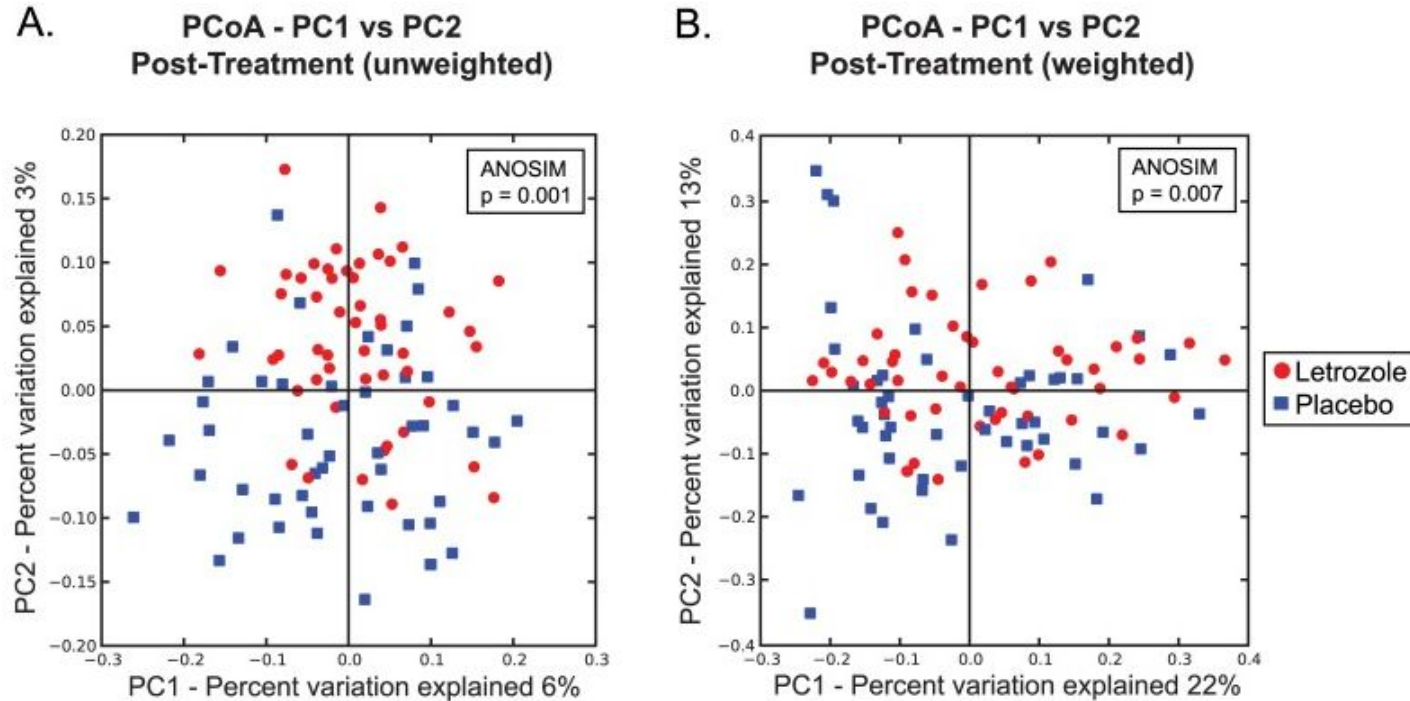
Which distance metric, for which question

Metric	Uses abundance?	Weights rare taxa	Good default when...
Jaccard	No, presence/absence only	Heavily (every taxon counts equally)	You care which taxa are present or absent, not how much
Bray-Curtis	Yes, abundance-weighted	Lightly	You care about shifts in the dominant community structure
UniFrac (unweighted)	No, presence/absence + tree	Heavily	You care which lineages are present, using phylogeny
UniFrac (weighted)	Yes, abundance + tree	Lightly	You care about abundance-weighted phylogenetic shifts

PERMANOVA doesn't care which distance matrix it receives, only that the choice matches your question.

ANOSIM

Nonparametric test which operates on a ranked dissimilarity matrix.



The null hypothesis is that the similarities within sites are smaller or equal to the similarities between sites.

Test the association of microbiome composition between treatments and among time points within treatments using weighted and unweighted UniFrac distances. Kelley et al., 2016. PloS one, 11(1)

Compositional data analysis

Don't want to throw away data?

- Don't rarefy
- Apply CLR (centre log ratio) transformation
 - Balances all features, avoids dominance by high-abundance taxa
 - Requires the addition of “pseudocounts”, zeros are converted to some “small” number.

Purpose-built methods for compositional data

- 1. Purpose-built methods model the compositional structure directly instead of testing raw relative abundances one taxon at a time, the mechanics are beyond today's scope, but know they exist and why they're preferred.
- 2. The more important choice is using one of them at all, instead of a naive per-taxon test.

Love, Huber & Anders (2014), Genome Biology — DESeq2;
Robinson, McCarthy & Smyth (2010), Bioinformatics — edgeR;
Fernandes et al. (2014), Microbiome — ALDEx2.

Operation	Standard approach	Compositional approach
Normalization	Rarefaction 'DESeq'	Centered Log Ratio (CLR)
Distance	Bray-Curtis UniFrac Jenson-Shanon	Aitchison
Ordination	PCoA (Abundance)	PCA (Variance)
Multivariate comparison	perMANOVA ANOSIM	perMANOVA ANOSIM
Correlation	Pearson Spearman	SparCC SpiecEasi
Differential abundance	metagenomSeq LEfSe DESeq	ALDEx2 ANCOM

Converting relative abundance to absolute abundance

1. Quantify the number of cells or amount of DNA for each sample
 - qPCR
 - Flow cytometry
 - Spike in known number of cells or DNA
2. Multiply each sample by their quantity to obtain absolute numbers and avoid compositionality.

Received: 19 July 2019 | Revised: 15 November 2019 | Accepted: 15 November 2019
DOI: 10.1002/mbe3.977

ORIGINAL ARTICLE

MicrobiologyOpen WILEY

Absolute quantitation of microbes using 16S rRNA gene metabarcoding: A rapid normalization of relative abundances by quantitative PCR targeting a 16S rRNA gene spike-in standard

Olivier Zemb¹ | Caroline S. Achard² | Jerome Hamelin³ | Marie-Léa De Almeida¹ |
Béatrice Gabinaud¹ | Laurent Cauquil¹ | Lisanne M.G. Verschuren^{4,5,6} |
Jean-Jacques Godon³

One test at $\alpha = 0.05$ is safe. A thousand tests is not.

	Family wise error (FWER)	False discovery rate (FDR)
Controls	P(at least one false positive), anywhere	Expected proportion of false positives among significant results
Correction	Bonferroni (or Holm, a less conservative stepwise version)	Benjamini-Hochberg
Best when	One false positive would be a serious problem (e.g. a confirmatory claim)	You expect many true effects and can tolerate a small, known fraction of false leads
Trade-off	Conservative, real effects get missed as m grows	Sacrifices some FWER control for much greater power

Exploratory 16S / shotgun screens of hundreds of taxa are almost always FDR territory. Benjamini-Hochberg dominates this field.

Bonferroni controls FWER; Benjamini-Hochberg controls FDR

BONFERRONI · FWER · CONSERVATIVE

$$\alpha_{adjusted} = \alpha / m$$

Guarantees $\text{FWER} \leq \alpha$. Simple, but extremely conservative when m is large, real effects get missed.

BENJAMINI-HOCHBERG · FDR · THE OMICS STANDARD

$$\text{Reject } p_{(1)} \dots p_{(k)}, k = \max i : p_{(i)} \leq (i/m) \cdot \alpha$$

Controls expected FDR at α . More power than Bonferroni, the standard choice for omics-scale testing.

Benjamini & Hochberg (1995), J R Stat Soc B, the original FDR paper, one of the most cited papers in statistics.

Applying Benjamini-Hochberg to 10 taxa

Rank p-values smallest to largest, compare each to its own threshold, and find the last one that still passes.

Rank (i)	Taxon	p-value	BH threshold ($i/10 \times 0.05$)	Result
1	Taxon 1	0.001	0.005	Significant
2	Taxon 2	0.004	0.010	Significant
3	Taxon 3	0.006	0.015	Significant
4	Taxon 4	0.011	0.020	Significant
5	Taxon 5	0.019	0.025	Significant
6	Taxon 6	0.028	0.030	Significant
7	Taxon 7	0.045	0.035	Not significant
8	Taxon 8	0.060	0.040	Not significant
9	Taxon 9	0.130	0.045	Not significant
10	Taxon 10	0.290	0.050	Not significant

Statistically significant is not the same as biologically important

1. A p-value says nothing about how big or meaningful the difference is.
2. With enough samples, even a practically irrelevant difference becomes 'statistically significant'.
3. Effect size answers the question a p-value can't: how large is the difference?
4. Always report both the p-value and effect size.

Large effect + Significant

Real and important

Small effect + Significant

Statistically, not practically important

Large effect + Not significant

Likely underpowered

Small effect + Not significant

Nothing here

The exact same tiny effect, at different sample sizes

Fixed throughout: a 0.3-percentage-point difference in relative abundance, $d = 0.15$ (most people would call this irrelevant).

n per group	p-value	Significant at $\alpha = 0.05$?
10	0.74	No
50	0.45	No
100	0.29	No
500	0.018	Yes
1,000	0.0008	Yes

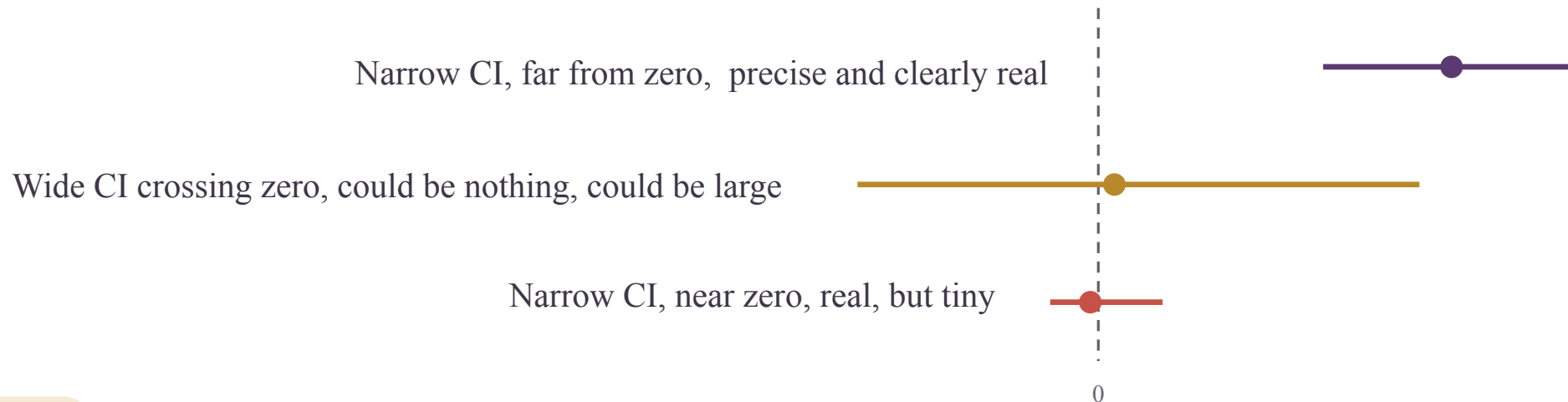
Push n to 5,000 and the same effect reaches $p < 1 \times 10^{-13}$

Effect size measures

Measure	Use it for	Rough interpretation
Cohen's d / Hedges' g	Comparing means of a continuous variable (diversity index, metabolite level)	0.2 small, 0.5 medium, 0.8 large
Log2 fold change	Per-taxon differential abundance	LFC = 1 → taxon is 2× more abundant
PERMANOVA R ²	Community-level (beta diversity) comparisons	Share of total community variance explained by the factor
Cliff's delta / rank-biserial r	Non-parametric comparisons (Wilcoxon / Mann-Whitney)	0.11 small, 0.28 medium, 0.43 large
Odds ratio / risk ratio	Presence/absence or binary outcomes	1 = no association; different from 1 = significant

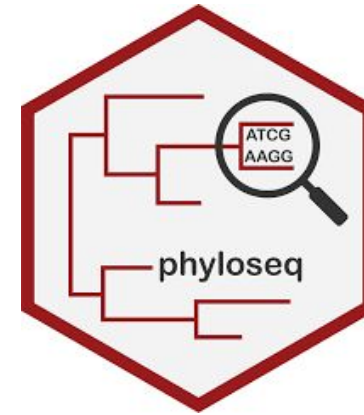
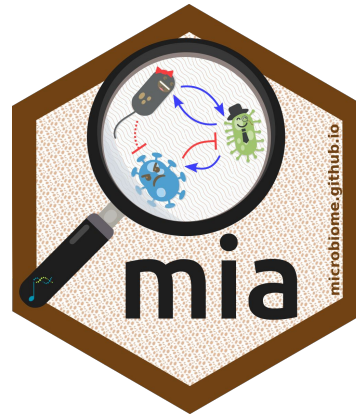
A confidence interval does more work than a p-value

1. A 95% CI on your effect size shows both its magnitude and your uncertainty about it, a p-value collapses both into a single yes/no.
2. A narrow CI far from zero and a wide CI that barely excludes zero can share the same p-value.
3. For multiple comparisons, adjust the CIs too, an unadjusted CI is just as optimistic as an unadjusted p-value.



Software to implement the analysis

1. mia
2. phyloseq
3. Vegan
4. biom
5. DESeq
6. DESeq2
7. metagenomeSeq
8. ANCOM
9. ALDEx and ALDEx2





1. Check normality and equal variance before defaulting to a t-test or ANOVA, microbiome data usually fails both.
2. Alpha diversity: rank-based tests or mixed models for repeated measures. Beta diversity: PERMANOVA on the distance matrix, ordination shows the pattern, it doesn't test it.
3. Differential abundance needs compositional-aware tools (ALDEx2, ANCOM-BC), not per-taxon t-tests, and not rarefying.
4. Correct for multiple comparisons, and know whether you're controlling FWER (Bonferroni) or FDR (Benjamini-Hochberg).
5. A p-value and an effect size answer different questions, report both, every time, even after correction.

Resources/Courses

- **Fundamentals of Biostatistics with R**
<https://github.com/e-ibrahimi/Fundamentals-of-Biostatistics-with-R->
- **Generalized linear models with R**
<https://github.com/e-ibrahimi/Generalized-Linear-Models-with-R>
- **Beginner Microbiome Analysis**
https://bioinformaticsdotca.github.io/BMB_2025/
- **Advance Microbiome Analysis**
https://bioinformaticsdotca.github.io/AMB_2025/
- **Orchestrating microbiome analysis with Bioconductor**
<https://microbiome.github.io/OMA/resources.html#data-containers-1>
- Ibrahim, E., Woford, B.N. **Ten essential tips for robust statistics in cell biology.** *Nat Cell Biol* **27**, 1884–1886 (2025). <https://doi.org/10.1038/s41556-025-01801-y>

Questions?
